Reeh–Schlieder and Bisognano–Wichmann theorems

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Abstract

The Reeh–Schlieder and Bisognano–Wichmann theorems reveal deep structures in the vacuum entanglement of a quantum field theory formulated in terms of Wightman axioms. This note discusses the content and proof of the two theorems.

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Conventions

We use the mostly positive metric $(-,+,+,+)$, $a\cdot b=-a^0b^0+\mathbf{a}\cdot\mathbf{b}$. The right and left wedges in the spacetime are defined as

$$W=\{x^1>|x^0|\},\qquad W'=-W=\{x^1<-|x^0|\}. \tag{1}$$

They are the causal complement of each other. The boost transformation is

$$\Lambda(s):(x^0,x^1,x^2,x^3)\mapsto(\cosh sx^0+\sinh sx^1,\sinh sx^0+\cosh sx^1,x^2,x^3), \tag{2}$$

with the generator K such that $U(\Lambda(s))=e^{isK}$ (for U in Definition 1.1).

1 The Reeh–Schlieder theorem

1.1 The Wightman axioms

Definition 1.1 (Bosonic QFT)**.** *A* bosonic quantum field theory *is characterized by a Hilbert space $\mathcal{H}$ with a positive-definite inner product, a dense subspace $\mathcal{D} \subset \mathcal{H}$, and the* field operators *$\varphi_a(f)$ for each Schwartz function $f \in \mathcal{S}(\mathbb{R}^4)$ satisfying the following axioms.*

(W1) *$\varphi_a(f)\mathcal{D} \subset \mathcal{D}$, and for each $\psi, \chi \in \mathcal{D}$ the map $f \mapsto \langle \psi, \varphi_a(f)\chi\rangle$ is continuous. We call the field φ_a* Hermitian *if $\varphi_a(\bar{f}) \subset \varphi_a(f)^\dagger$.*

(W2) *The proper orthochronous Poincaré group $\mathcal{P}_+^\uparrow$ has a strongly continuous unitary representation $(a,\Lambda) \mapsto U(a,\Lambda)$ on $\mathcal{H}$, such that for a scalar field φ,*

$$U(a,\Lambda)\,\varphi(f)\,U(a,\Lambda)^{-1} = \varphi\big(f \circ (a,\Lambda)^{-1}\big), \tag{3}$$

where $\big(f \circ (a,\Lambda)^{-1}\big)(x) = f\big(\Lambda^{-1}(x-a)\big)$. In terms of the point fields of Remark 1.2 below, this reads $U(a,\Lambda)\varphi(x)U(a,\Lambda)^{-1} = \varphi(\Lambda x + a)$. Furthermore, there is a unique one-dimensional invariant subspace, spanned by a unit vector Ω called the vacuum *($\langle\Omega|\Omega\rangle = 1$).*

(W3) *The translations $a \mapsto U(a,1)$ form a strongly continuous unitary representation of $\mathbb{R}^4$. By Stone's theorem it has commuting self-adjoint generators P^μ such that $U(a,1) = \exp[-iP\cdot a]$. The joint spectrum $\operatorname{spec} P$ of the generators P^μ is included in the closed forward lightcone $\overline{V}_+ = \{\, p \,|\, p^2 \le 0,\ p^0 \ge 0 \,\}$.*

(W4) *$[\varphi_a(f), \varphi_b(g)] = 0$ whenever the supports of f, g are spacelike-separated:*

$$x \in \operatorname{supp} f,\ y \in \operatorname{supp} g \quad\Longrightarrow\quad (x-y)^2 > 0. \tag{4}$$

(W5) *For an open set $O \subset \mathbb{R}^4$, let $\mathcal{P}(O)$ be the $*$-algebra generated by 1 and the smeared fields $\varphi_a(f)$ with $f \in C_c^\infty(\mathbb{R}^4)$, $\operatorname{supp} f \subset O$. The vacuum $|\Omega\rangle$ is* cyclic *in the sense that $\mathcal{P}(\mathbb{R}^4)|\Omega\rangle$ is dense in $\mathcal{H}$. (Streater–Wightman allow $f \in \mathcal{S}(\mathbb{R}^4)$ here; the two versions are equivalent by the continuity in (W1).)*

Remark 1.2 (Point fields)**.** *Formally $\varphi_a(f) = \int \mathrm{d}^4x\, f(x)\,\varphi_a(x)$, so that $\varphi_a(x)$ "$= \varphi_a(\delta_x)$" is obtained by inserting the delta function δ_x, which is not a Schwartz function. In this note we compute with the point fields $\varphi_a(x)$ as physicists do. Each identity derived this way is an identity between tempered distributions, and becomes a rigorous statement after smearing all point fields with test functions; we comment on this where it matters.*

Remark 1.3 (Fermions)**.** *Fermionic fields can be included by replacing $\mathcal{P}_+^\uparrow$ in (W2) with its double cover $\mathbb{R}^4 \rtimes SL(2,\mathbb{C})$, under which spinor fields transform as $U(a,A)\psi(x)U(a,A)^{-1} = S(A^{-1})\psi(\Lambda(A)x+a)$, and by replacing (W4) with graded locality: two fermionic fields anticommute, and all other pairs commute, at spacelike separation. The fermion parity $\Gamma = U(0,-1)$ fixes Ω. None of the arguments in this note requires this generality.*

1.2 Implications of the axiom

The spectrum condition (W3) consists of two inequalities on the joint spectrum of $P^\mu = (H, \mathbf{P})$:

$$H \geq 0, \qquad |\mathbf{P}| \leq H. \tag{5}$$

Each inequality makes the dependence of matrix elements on translations analytic in one direction: the first in time, the second in space. This is the only input of the Reeh–Schlieder theorem.

Proposition 1.4 (Spectral analyticity)**.** *Let $\Psi, \Phi \in \mathcal{H}$ and write $U(x) = U(x,1) = e^{-iP\cdot x} = e^{iHt}e^{-i\mathbf{P}\cdot\mathbf{a}}$ for $x = (t, \mathbf{a})$.*

(i) (Energy is bounded below.) *For fixed $\mathbf{a}$, the function $t \mapsto \langle\Psi|e^{iHt}e^{-i\mathbf{P}\cdot\mathbf{a}}|\Phi\rangle$ extends continuously to $\operatorname{Im} t \geq 0$ and holomorphically to $\operatorname{Im} t > 0$, with modulus at most $\|\Psi\|\|\Phi\|$.*

(ii) (Momentum is bounded by energy.) *If χ is a bounded function supported in $(-\infty, \Lambda]$, the function $\mathbf{a} \mapsto \langle\Psi|e^{-i\mathbf{P}\cdot\mathbf{a}}\chi(H)|\Phi\rangle$ extends to an entire function on $\mathbb{C}^3$.*

Proof. (i) By the spectral theorem the function is $\int e^{iEt}\,\mathrm{d}\mu(E,\mathbf{p})$ for the complex measure $\mathrm{d}\mu = \langle\Psi|\mathrm{d}E_P\, e^{-i\mathbf{P}\cdot\mathbf{a}}|\Phi\rangle$ supported in $\overline{V}_+$. For $E \geq 0$ and $\operatorname{Im} t \geq 0$ we have $|e^{iEt}| = e^{-E\operatorname{Im} t} \leq 1$, so the integral converges and is continuous by dominated convergence. For $\operatorname{Im} t \geq \eta > 0$ the derivative iEe^{iEt} is bounded by $(e\eta)^{-1}$, so one may differentiate under the integral.

(ii) The vector $\chi(H)\Phi$ lies in the spectral subspace of P for the set $\overline{V}_+ \cap \{p^0 \leq \Lambda\}$, on which $|\mathbf{p}| \leq p^0 \leq \Lambda$ by Eq. (5). On this subspace each P^j is bounded with norm at most Λ, so the exponential series of $e^{-i\mathbf{P}\cdot\mathbf{a}}$ converges in norm for every $\mathbf{a} \in \mathbb{C}^3$. □

The two parts combine into a unique continuation property: a matrix element of the translation operator cannot vanish near the origin without vanishing everywhere.

Lemma 1.5 (Unique continuation along translations)**.** *Let $\Psi, \Phi \in \mathcal{H}$ and $f(x) = \langle\Psi|U(x)|\Phi\rangle$. If f vanishes on a cylinder $N = (-\epsilon, \epsilon)_t \times B_\delta(\mathbf{0})$, then $f \equiv 0$ on $\mathbb{R}^4$.*

Proof. Step 1: time, from $H \geq 0$. Fix $|\mathbf{a}| < \delta$. By Proposition 1.4(i), $t \mapsto f(t, \mathbf{a})$ is holomorphic in the upper half-plane, continuous up to the real axis, and zero on $(-\epsilon, \epsilon)$. Setting $f(t, \mathbf{a}) := \overline{f(\bar{t}, \mathbf{a})}$ for $\operatorname{Im} t < 0$ gives, by the Schwarz reflection principle, a function holomorphic in a neighborhood of $(-\epsilon, \epsilon)$ that vanishes on this interval. Hence it vanishes in the whole upper half-plane, and by continuity $f(t, \mathbf{a}) = 0$ for all $t \in \mathbb{R}$.

Step 2: space, from $|\mathbf{P}| \leq H$. Let $\chi \in C_c^\infty(\mathbb{R})$ and $g(t) = \frac{1}{2\pi}\int \mathrm{d}E\, \chi(E)e^{-iEt}$, so that $\int \mathrm{d}t\, g(t)e^{iHt} = \chi(H)$. For $|\mathbf{a}| < \delta$,

$$\langle\Psi|e^{-i\mathbf{P}\cdot\mathbf{a}}\chi(H)|\Phi\rangle = \int \mathrm{d}t\, g(t)\, f(t, \mathbf{a}) = 0. \tag{6}$$

Here Step 1 is essential: since χ has compact support in energy, g cannot have compact support in time, and the integral involves $f(t, \mathbf{a})$ at all times. By Proposition 1.4(ii) the left-hand side is entire in $\mathbf{a}$, so it vanishes for all $\mathbf{a} \in \mathbb{R}^3$. Replacing $\chi(E)$ by $\chi(E)e^{iEt}$ gives $\langle\Psi|e^{-i\mathbf{P}\cdot\mathbf{a}}\chi(H)e^{iHt}|\Phi\rangle = 0$ for all $(t, \mathbf{a})$. Finally take $\chi_n \to 1$ pointwise with $|\chi_n| \leq 1$; then $\chi_n(H) \to 1$ strongly and $f \equiv 0$. □

1.3 The theorem

Recall the algebras $\mathcal{P}(O)$ of (W5).

Theorem 1.6 (Reeh–Schlieder)**.** *In a Wightman QFT, $\mathcal{P}(O)\Omega$ is dense in $\mathcal{H}$ for every nonempty open $O \subset \mathbb{R}^4$.*

Proof. Let $\Psi \perp \mathcal{P}(O)\Omega$. By (W5) it suffices to show $\Psi \perp \mathcal{P}(\mathbb{R}^4)\Omega$. The whole proof is Lemma 1.5 applied to one translation variable at a time.

Step 1: reduction to translates of a small region. Choose an open set O_1 and a cylinder $N = (-\epsilon, \epsilon)_t \times B_\delta(\mathbf{0})$ such that $O_1 + x \subset O$ for all $x \in N$. The test functions in $\mathcal{P}(\mathbb{R}^4)$ have compact support, so a partition of unity writes each f_j as a finite sum of test functions supported in translates $O_1 + x$. Hence $\mathcal{P}(\mathbb{R}^4)\Omega$ is spanned by vectors of the form

$$U(x_1)A_1U(x_1)^{-1}\,U(x_2)A_2U(x_2)^{-1}\cdots U(x_n)A_nU(x_n)^{-1}\,\Omega, \qquad A_j \in \mathcal{P}(O_1),\ x_j \in \mathbb{R}^4. \tag{7}$$

Using $U(x)\Omega = \Omega$ and the difference variables $y_1 = x_1$, $y_j = x_j - x_{j-1}$, its overlap with Ψ is

$$F(y_1, \dots, y_n) = \langle \Psi | U(y_1)A_1\,U(y_2)A_2 \cdots U(y_n)A_n|\Omega\rangle. \tag{8}$$

We must show $F \equiv 0$.

Step 2: vanishing near the origin. Let $N_0 = (-\epsilon/n, \epsilon/n)_t \times B_{\delta/n}(\mathbf{0})$. If all $y_j \in N_0$, then every $x_j = y_1 + \cdots + y_j$ lies in N, each factor $U(x_j)A_jU(x_j)^{-1}$ belongs to $\mathcal{P}(O)$, and $F = 0$ by assumption.

Step 3: spreading, one variable at a time. For $k = 0, \dots, n$ let (C_k) be the statement: $F(y) = 0$ whenever $y_1, \dots, y_k \in \mathbb{R}^4$ are arbitrary and $y_{k+1}, \dots, y_n \in N_0$. We have just shown (C_0). Assume (C_{k-1}) and fix $y_1, \dots, y_{k-1} \in \mathbb{R}^4$ and $y_{k+1}, \dots, y_n \in N_0$. As a function of y_k,

$$\begin{gathered} F = \langle \Psi_k | U(y_k) | \Phi_k \rangle, \\ \langle \Psi_k | = \langle \Psi | U(y_1)A_1 \cdots U(y_{k-1})A_{k-1}, \qquad |\Phi_k\rangle = A_kU(y_{k+1})A_{k+1}\cdots U(y_n)A_n|\Omega\rangle, \end{gathered} \tag{9}$$

and (C_{k-1}) says that it vanishes for $y_k \in N_0$. (Moving the operators $U(y_j)A_j$ onto $\langle\Psi|$ is literally valid for bounded A_j; for field polynomials the same step is carried out at the level of the distribution $\langle\Psi|\varphi(x_1)\cdots\varphi(x_n)|\Omega\rangle$, cf. Remark 1.2.) Lemma 1.5 gives $F = 0$ for all $y_k \in \mathbb{R}^4$, which is (C_k). Then (C_n) is the claim, and $\Psi \perp \mathcal{P}(\mathbb{R}^4)\Omega$. □

Remark 1.7 (Relation to Euclidean field theory)**.** *The vacuum has a Euclidean path integral representation over the lower half-space $\mathbb{R}_{\tau\le 0} \times \mathbb{R}^d$ (see Eq. (17)). Inserting field operators at $(\tau_j, \mathbf{x}_j)$ $(0 > \tau_1 > \tau_2 > \cdots)$, the vectors of the form*

$$\Phi_E = e^{H\tau_1}\phi(\mathbf{x}_1)e^{H(\tau_2-\tau_1)}\phi(\mathbf{x}_2)\cdots\Omega \tag{10}$$

are precisely the continuation of $\phi(x_1)\cdots\phi(x_n)\Omega$ along imaginary time $x_j^0 \to i|\tau_j|$, in accordance with Proposition 1.4. Osterwalder–Schrader system reconstructs the Hilbert space by these vectors. Thus, these vectors are dense in $\mathcal{H}$.

Corollary 1.8. *If O is an open set whose causal complement O' has nonempty interior, Ω is separating for $\mathcal{P}(O)$ in the sense that $A \in \mathcal{P}(O)$ and $A\Omega = 0$ imply $A = 0$ on $\mathcal{D}$.*

Proof. Let $\Psi \in \mathcal{D}$ and $B \in \mathcal{P}(O')$. Writing $A^* \in \mathcal{P}(O)$ for the formal adjoint of A (reverse the order of the factors and replace f by $\bar{f}$; by (W1), $\langle \Phi | A | \Psi \rangle = \langle A^* \Phi | \Psi \rangle$ for $\Phi, \Psi \in \mathcal{D}$), locality (W4) gives

$$\langle A^* \Psi | B\Omega \rangle = \langle \Psi | AB | \Omega \rangle = \langle \Psi | BA | \Omega \rangle = 0.$$

Since $\mathcal{P}(O')\Omega$ is dense by Theorem 1.6, $A^*\Psi = 0$. Then $\langle \Phi | A | \Psi \rangle = \langle A^* \Phi | \Psi \rangle = 0$ for all $\Phi, \Psi \in \mathcal{D}$, so $A = 0$ on $\mathcal{D}$. (For an unbounded A, vanishing on the dense set $\mathcal{P}(O')\Omega$ alone would not suffice; this is why we pass through A^*. For a bounded local operator the first line already finishes the proof.) □

2 Introduction to modular theory

2.1 Finite dimensional model

Let $\mathcal{H} = \mathbb{C}^n \otimes \mathbb{C}^n$, $M = M_n(\mathbb{C}) \otimes 1$, $\Omega = \sum_i \sqrt{p_i} |i\rangle |i\rangle$ $(p_i > 0)$, $\rho = \mathrm{diag}(p_i)$.

Tomita operator. Define an anti-linear operator S by $SA\Omega = A^\dagger \Omega$ $(A \in M)$. From the polar decomposition $S = J\Delta^{1/2}$ where $\Delta = S^\dagger S$, one obtains the relations

$$\Delta = \rho \otimes \rho^{-1}, \qquad J\left(\sum c_{ij} |ij\rangle\right) = \sum \overline{c_{ij}} |ji\rangle. \tag{11}$$

Indeed, $(E_{ij} \otimes 1)\Omega = \sqrt{p_j} |ij\rangle$ for $E_{ij} = |i\rangle\langle j|$, so $S|ij\rangle = \sqrt{p_i/p_j}\, |ji\rangle$ and $S^\dagger |kl\rangle = \sqrt{p_l/p_k}\, |lk\rangle$ (with $\langle S\psi | \chi \rangle = \langle S^\dagger \chi | \psi \rangle$ for anti-linear operators). Hence $\Delta |ij\rangle = (p_i/p_j)|ij\rangle$ and $J|ij\rangle = S\Delta^{-1/2}|ij\rangle = |ji\rangle$. Writing $\rho = e^{-H_A}$, the modular operator $\Delta = e^{-H_A} \otimes e^{+H_B}$ contains the entanglement Hamiltonian of each factor with opposite signs. Under $\mathcal{H} \cong M_n(\mathbb{C})$, $|ij\rangle \leftrightarrow E_{ij}$, one has $\Omega \leftrightarrow \rho^{1/2}$, $\Delta(X) = \rho X \rho^{-1}$ and $J(X) = X^\dagger$, so that JMJ acts by right multiplication and equals M'.

KMS condition. For a Gibbs state $\langle X \rangle = \mathrm{tr}(e^{-\beta H} X)/Z$ and $\alpha_t(X) = e^{iHt} X e^{-iHt}$, cyclicity of the trace gives

$$\langle A\, \alpha_{t+i\beta}(B) \rangle = \langle \alpha_t(B)\, A \rangle. \tag{12}$$

This formulation does not refer to a density matrix, and is taken as the definition of a thermal state at inverse temperature β in infinite dimensions. Define the *modular Hamiltonian* $K_{\mathrm{mod}} := -\log \Delta$ and the *modular flow* $\tau_s(A) := \Delta^{-is} A \Delta^{is} = e^{isK_{\mathrm{mod}}} A e^{-isK_{\mathrm{mod}}}$. In the model, $K_{\mathrm{mod}} = H_A \otimes 1 - 1 \otimes H_B$ and $\tau_s(A \otimes 1) = e^{isH_A} A e^{-isH_A} \otimes 1$ is the time evolution generated by H_A, so Ω satisfies Eq. (12) with $\beta = 1$.

2.2 Tomita–Takesaki theorem

Let $\mathcal{R} \subset B(\mathcal{H})$ be a von Neumann algebra and Ω be a cyclic separating vector of $\mathcal{R}$. Define $S_0 A\Omega = A^\dagger\Omega$ on $\mathcal{R}\Omega$ and $F_0 B'\Omega = B'^\dagger\Omega$ on $\mathcal{R}'\Omega$. These are well defined because Ω is separating for $\mathcal{R}$ and for $\mathcal{R}'$ (a vector is cyclic for $\mathcal{R}$ if and only if it is separating for $\mathcal{R}'$). The computation

$$\langle S_0 A\Omega | B'\Omega\rangle = \langle\Omega|AB'|\Omega\rangle = \langle\Omega|B'A|\Omega\rangle = \langle F_0 B'\Omega | A\Omega\rangle$$

shows $F_0 \subset S_0^\dagger$, so $S_0^\dagger$ is densely defined and S_0 is closable. Its closure S is the *Tomita operator*, and the polar decomposition

$$S = J\Delta^{1/2}, \qquad \Delta := S^\dagger S \tag{13}$$

defines the *modular operator* Δ (positive, self-adjoint, injective) and the *modular conjugation* J (anti-unitary). From $S = S^{-1}$ and uniqueness of the polar decomposition one finds $J^2 = 1$, $J\Delta J = \Delta^{-1}$, $\Delta\Omega = \Omega$, $J\Omega = \Omega$.

Theorem 2.1 (Tomita–Takesaki)**.** *$\Delta^{it}\mathcal{R}\Delta^{-it} = \mathcal{R}$ for all $t \in \mathbb{R}$, and $J\mathcal{R}J = \mathcal{R}'$.*

We quote this theorem without proof. In the finite-dimensional model both statements can be checked by hand.

Proposition 2.2 (KMS)**.** *For $A, B \in \mathcal{R}$, the function $F(s) = \langle\Omega|A\,\tau_s(B)|\Omega\rangle$ extends continuously to $0 \le \operatorname{Im} s \le 1$, holomorphically inside, and $F(s+i) = \langle\Omega|\tau_s(B)\,A|\Omega\rangle$.*

Proof. Since $\Delta^{is}\Omega = \Omega$, $F(s) = \langle A^\dagger\Omega | \Delta^{-is} B\Omega\rangle$ with $A^\dagger\Omega, B\Omega \in D(S) = D(\Delta^{1/2})$, so the spectral representation of Δ continues F to the strip with $F(u+iy) = \langle \Delta^{y/2}A^\dagger\Omega | \Delta^{-iu}\Delta^{y/2}B\Omega\rangle$. At $y = 1$, using $\Delta^{1/2}A^\dagger\Omega = JA\Omega$, $\Delta^{1/2}B\Omega = JB^\dagger\Omega$, $J\Delta^{iu} = \Delta^{iu}J$ and anti-unitarity,

$$F(u+i) = \langle JA\Omega | J\Delta^{-iu}B^\dagger\Omega\rangle = \langle \Delta^{-iu}B^\dagger\Omega | A\Omega\rangle = \langle\Omega|\tau_u(B)\,A|\Omega\rangle. \tag{14}$$

The mathematical literature uses $\sigma_t = \operatorname{Ad}\Delta^{it} = \tau_{-t}$, for which the same statement reads as the KMS condition at $\beta = -1$. □

Comparing with Eq. (12), Ω is a thermal state at $\beta = 1$ for the modular Hamiltonian $K_{\mathrm{mod}} = -\log\Delta$. When $\mathcal{R}$ is of type III, as are local algebras in quantum field theory, neither ρ nor H_A exists, but Δ does; K_{mod} is the rigorous counterpart of the entanglement Hamiltonian.

3 The Bisognano–Wichmann theorem

3.1 Statement

Let $\mathcal{R}(W)$ be the von Neumann algebra generated by the fields smeared in the wedge W.[1] By Theorem 1.6 and its corollary, together with the condition in the footnote, Ω is cyclic and

[1] For unbounded fields this requires a technical condition, e.g. that the smeared fields are locally essentially self-adjoint, so that their bounded functions generate a von Neumann algebra whose action on Ω reproduces that of $\mathcal{P}(W)$ [1]. For the free field one uses Weyl operators directly.

separating for $\mathcal{R}(W)$, so the modular objects Δ, J of Section 2 are defined. Let Θ be the antiunitary CPT operator, $\Theta\varphi(x)\Theta^{-1} = \varphi(-x)$, and R_1 the rotation by π in the (x^2, x^3) plane.

Theorem 3.1 (Bisognano–Wichmann)**.** *The modular operator and modular conjugation of* $(\mathcal{R}(W), \Omega)$ *are*

$$\Delta = e^{-2\pi K}, \qquad J = \Theta\, U(R_1), \tag{15}$$

i.e. $K_{\mathrm{mod}} = 2\pi K$*, so that the modular flow is the boost,* $\tau_s = \mathrm{Ad}\, U(\Lambda(2\pi s))$*, and* $J\varphi(x)J = \varphi(-x^0, -x^1, x^2, x^3)$ *for a scalar field.*

Below (§3.2) we give the Euclidean path integral argument, which is formal but makes every ingredient visible. The rigorous proof [1, 2] replaces the path integral by the analytic continuation of $e^{isK}A\Omega$ ($A \in \mathcal{P}(W)$) to $s = i\pi$, which is possible by the spectrum condition because $\mathrm{Im}\,\Lambda(iy)x \in \overline{V}_+$ for $x \in W$ and $0 \le y \le \pi$; the endpoint $\Lambda(i\pi)x = (-x^0, -x^1, x^2, x^3)$ is again a real point.

Corollary 3.2.

(i) (KMS) *The boosts* $\alpha_s = \mathrm{Ad}\, e^{isK}$ *(see Eq. (2)) satisfy the KMS condition at* $\beta = 2\pi$ *on* $\mathcal{R}(W)$*: for* $A, B \in \mathcal{R}(W)$*,* $F(s) = \langle\Omega|A\,\alpha_s(B)|\Omega\rangle$ *extends to* $0 \le \mathrm{Im}\, s \le 2\pi$ *with* $F(s + 2\pi i) = \langle\Omega|\alpha_s(B)\,A|\Omega\rangle$*.*

(ii) (Wedge duality) $\mathcal{R}(W)' = \mathcal{R}(W')$*.*

Proof. (i) By Theorem 3.1, $\tau_s = \alpha_{2\pi s}$, which preserves $\mathcal{R}(W)$ because boosts map W to itself. Proposition 2.2 applied to $\tau_{s/2\pi} = \alpha_s$ gives the claim. In the path integral of §3.2, the same statement is the single-valuedness of Euclidean correlation functions under $\vartheta \to \vartheta + 2\pi$, since ϑ is the imaginary boost parameter. (ii) By Theorem 2.1, $\mathcal{R}(W)' = J\mathcal{R}(W)J$. By Theorem 3.1, $J\varphi(x)J = \varphi(-x^0, -x^1, x^2, x^3)$ maps fields smeared in W to fields smeared in W', so $J\mathcal{R}(W)J = \mathcal{R}(W')$. □

Remark 3.3 (Unruh effect)**.** *Item (i) says that the vacuum restricted to* W *is thermal at inverse temperature* 2π *with respect to the boost. An observer on the orbit*

$$x(\tau) = a^{-1}(\sinh a\tau, \cosh a\tau, 0, 0) = \Lambda(a\tau)(0, a^{-1}, 0, 0)$$

evolves by $\alpha_{a\tau}$*, i.e. with Hamiltonian* aK *in proper time* τ*. The period* 2π *in* s *is the period* $2\pi/a$ *in* τ*, and therefore the observer sees the temperature* $T = a/2\pi$*.*

3.2 The path integral argument

Formal derivation. By §2.1, Δ and J are read off from the Schmidt decomposition of Ω with respect to $\mathcal{H}_R \otimes \mathcal{H}_L$: the Schmidt coefficients give Δ, and the pairing of left and right Schmidt vectors gives J. We compute this decomposition with the Euclidean path integral. We formally

write $\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R$, where $\mathcal{H}_L, \mathcal{H}_R$ are the Hilbert space at $\tau = 0$ associated to the subsystem $\{x^1 < 0\}$ and $\{x^1 > 0\}$, respectively.

Step 1: the vacuum functional.

$$\langle\varphi|e^{-HT}|\chi\rangle = \int_{\phi(-T)=\chi}^{\phi(0)=\varphi} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{16}$$

The limit $T \to \infty$ projects the state as $e^{-HT}|\chi\rangle \to e^{-E_0 T}|\Omega\rangle\langle\Omega|\chi\rangle$,

$$\Psi_0[\varphi] := \langle\varphi|\Omega\rangle = \mathcal{N} \int_{\substack{\tau<0 \\ \phi(0^-)=\varphi}} \mathcal{D}\phi\, e^{-S_E[\phi]}. \tag{17}$$

Step 2: the density matrix. If $\rho = |\Omega\rangle\langle\Omega|$, the matrix element $\langle\varphi|\rho|\varphi'\rangle = \Psi_0[\varphi]\overline{\Psi_0[\varphi']}$ is represented as a path integral over $\mathbb{R}^{d+1}\backslash\{\tau = 0\} = (\mathbb{R}_{\tau<0} \times \mathbb{R}^d) \cup (\mathbb{R}_{\tau>0} \times \mathbb{R}^d)$ subject to the boundary condition $\phi(\tau = 0^-) = \varphi,\ \phi(\tau = 0^+) = \varphi'$. Partial trace over $\mathcal{H}_L$ means gluing half of the hyperplane $\{\tau = 0,\ x^1 < 0\}$ back,

$$\langle\varphi_R|\rho_R|\varphi'_R\rangle = \frac{1}{Z} \int_{\substack{\phi(x^1>0,\,0^-)=\varphi_R \\ \phi(x^1>0,\,0^+)=\varphi'_R}} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{18}$$

where Z is the path integral over the whole $\mathbb{R}^{d+1}$.

Step 3: reparametrization. Reparametrize the (x^1, τ) plane by $x^1 = r\cos\vartheta,\ \tau = r\sin\vartheta$. The Euclidean metric becomes $\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\mathrm{d}\vartheta^2 + \mathrm{d}\mathbf{x}_\perp^2$, independent of ϑ, so ∂_ϑ is a Killing vector: ϑ is a static "time" with lapse r. The states in $\mathcal{H}_R = \mathcal{H}(\vartheta = 0)$ evolve along ϑ. The identification $\mathcal{H}(\vartheta) = U(\vartheta)\mathcal{H}(0) \simeq \mathcal{H}_R$ is canonical, since rotation in the (x^1, τ) plane is a symmetry of the Euclidean field theory. Hence, there is a well-defined (formally self-adjoint, unbounded) operator K_E on $\mathcal{H}_R$ with no dependence on ϑ, such that

$$\langle\varphi_R|\rho_R|\varphi'_R\rangle = \frac{1}{Z}\langle\varphi_R|e^{-2\pi K_E}|\varphi'_R\rangle. \tag{19}$$

Since rotation by $\delta\vartheta$ displaces $(x^1, 0)$ by $\delta\tau = x^1\delta\vartheta$,

$$K_E = \int_{x^1>0} \mathrm{d}^d x\, x^1 T^{00}(0, \mathbf{x}) =: K_R. \tag{20}$$

A small boost $\Lambda(s)$ moves $(0, x^1)$ by $\delta x^0 = s\, x^1$, the real-time translation with the same "lapse" x^1. Thus K_R is the boost generator restricted to the right half, and a Euclidean rotation by ϑ is a boost by imaginary rapidity; in particular the rotation by π, $(x^1, \tau) \mapsto (-x^1, -\tau)$, is $\Lambda(i\pi) : (x^0, x^1) \mapsto (-x^0, -x^1)$. The sign of the exponent is fixed by convergence, since $K_R \geq 0$ formally. Altogether,

$$\rho_R = \frac{e^{-2\pi K_R}}{Z}, \qquad Z = \operatorname{tr} e^{-2\pi K_R}. \tag{21}$$

Step 4: Schmidt decomposition of Ω (thermofield double). The lower half-space in Eq. (17) is the range $\pi \leq \vartheta \leq 2\pi$, bounded by the left ray ($\vartheta = \pi$, data φ_L) and the right ray ($\vartheta = 2\pi$, data φ_R). It is an angular evolution by π:

$$\Psi_0[\varphi_L, \varphi_R] \propto \langle\varphi_R|e^{-\pi K_R}|\Theta\varphi_L\rangle, \tag{22}$$

where the rotation by π identifies the left ray with the right one; since it reverses τ, it is antiunitary, and it is the operator $\Theta U(R_1)$ above. In an eigenbasis $K_R|n\rangle = \omega_n|n\rangle$,

$$|\Omega\rangle = \frac{1}{\sqrt{Z}} \sum_n e^{-\pi\omega_n} |n\rangle_R \otimes |\Theta n\rangle_L. \tag{23}$$

All Schmidt coefficients are positive, which is the formal content of the Reeh–Schlieder theorem for the wedge.

Step 5: the modular objects. Equation Eq. (23) is the finite-dimensional model of §2.1 with $p_n = e^{-2\pi\omega_n}/Z$. Defining $K_L = \Theta K_R \Theta$ and noting that boosts run backwards in time on the left, the full boost generator is $K = K_R \otimes 1 - 1 \otimes K_L$. The formulas of §2.1 then give

$$\Delta = \rho_R \otimes \rho_L^{-1} = e^{-2\pi K_R} \otimes e^{2\pi K_L} = e^{-2\pi K}, \tag{24a}$$

$$J\big(|m\rangle_R |\Theta n\rangle_L\big) = |n\rangle_R |\Theta m\rangle_L \text{ (coefficients conjugated)}, \tag{24b}$$

so J is the rotation by π itself. Equivalently, for A acting on $\mathcal{H}_R$ with $A_{mn} = \langle m|A|n\rangle$,

$$e^{-\pi K}(A \otimes 1)\Omega = \frac{1}{\sqrt{Z}} \sum_{m,n} e^{-\pi\omega_m} A_{mn} |m\rangle|\Theta n\rangle = J(A^\dagger \otimes 1)\Omega, \tag{25}$$

which is $S = J\Delta^{1/2}$ with $\Delta^{1/2} = e^{-\pi K}$. In the path integral this is the statement that rotating an insertion through the lower half-plane by the angle π carries A on the right ray to $A^\dagger$, read through Θ, on the left ray. □

Remark 3.4 (What is formal)**.** *(1) There is no factorization $\mathcal{H} = \mathcal{H}_L \otimes \mathcal{H}_R$ in a continuum theory: local algebras are of type III, and ρ_R, K_R do not exist separately; only Δ and K do. (2) The angular slicing degenerates at the tip $r = 0$, where the rotation has a fixed point. Removing a disc of radius ε regularizes it; as $\varepsilon \to 0$, $Z = \operatorname{tr} e^{-2\pi K_R}$ diverges, which is the area-law divergence of the entanglement entropy. (3) For fermions the argument is unchanged, except that a spinor picks up a sign under rotation by 2π; this sign supplies exactly the antiperiodic boundary condition of a fermionic thermal trace, consistent with $\rho_R \propto e^{-2\pi K_R}$.*

References

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